



VIT[®]

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(Deemed to be University under section 3 of UGC Act, 1956)

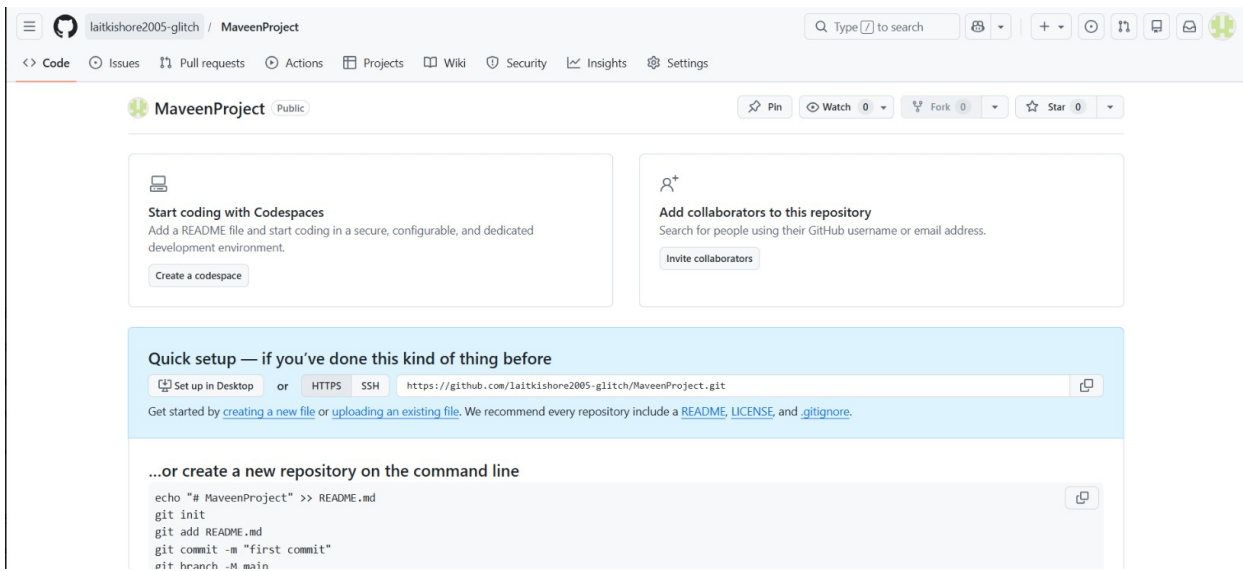
WINTER SEMESTER 2025-26

ASSIGNMENT - 3

FACULTY : SWETA BHATTACHARYA
NAME : SHARU SREE
REG NO : 23MIS0403
SUBJECT : AGILE DEVELOPMENT
PROCESS AND DEVOPS
COURSE CODE: ISWE406P

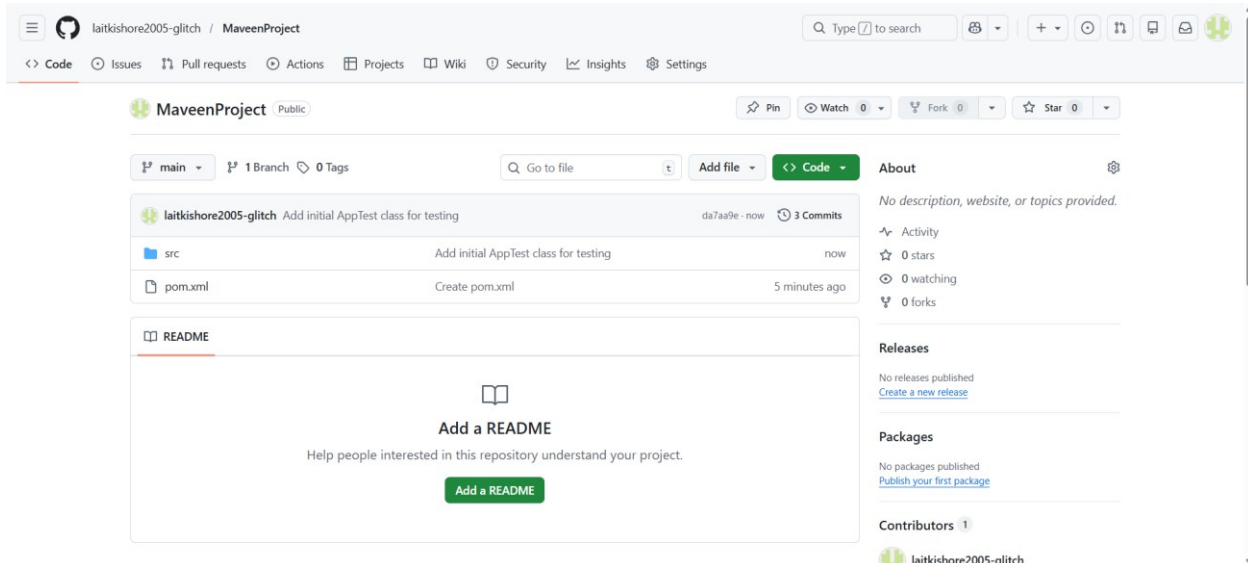
1. Create a Maven Project in Git

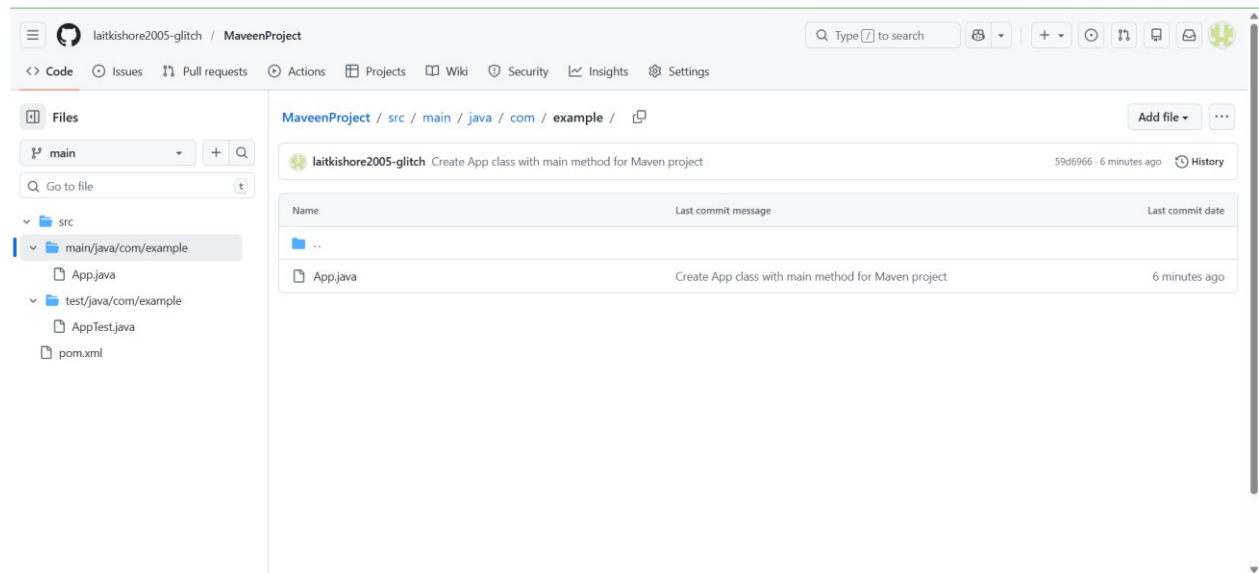
• Create a new Git repository



• Create a simple Maven project structure manually, including the following files and folders:

- i) pom.xml
- ii) src/main/java/com/example/App.java
- iii) src/test/java/com/example/AppTest.java

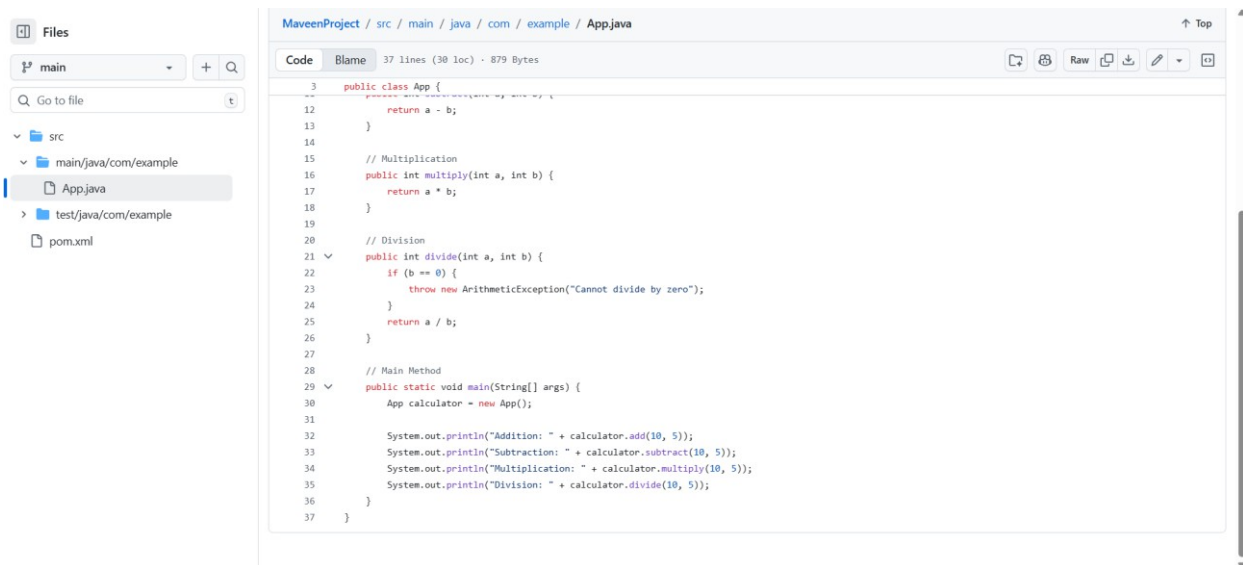




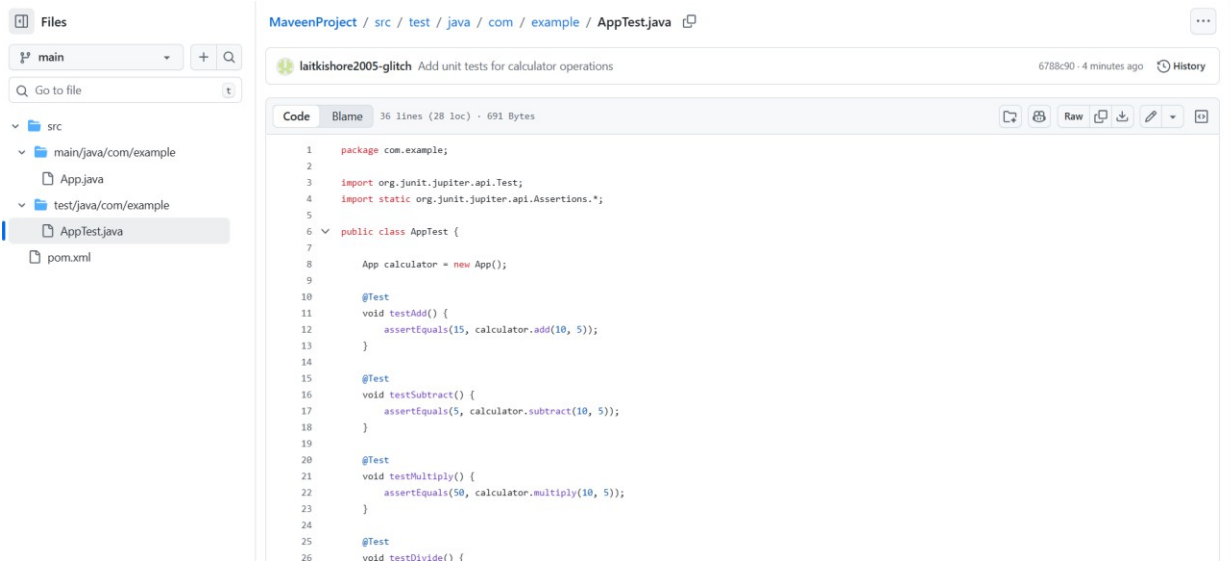
2. Implement the Java Program

- Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:

Simple Calculator Application:



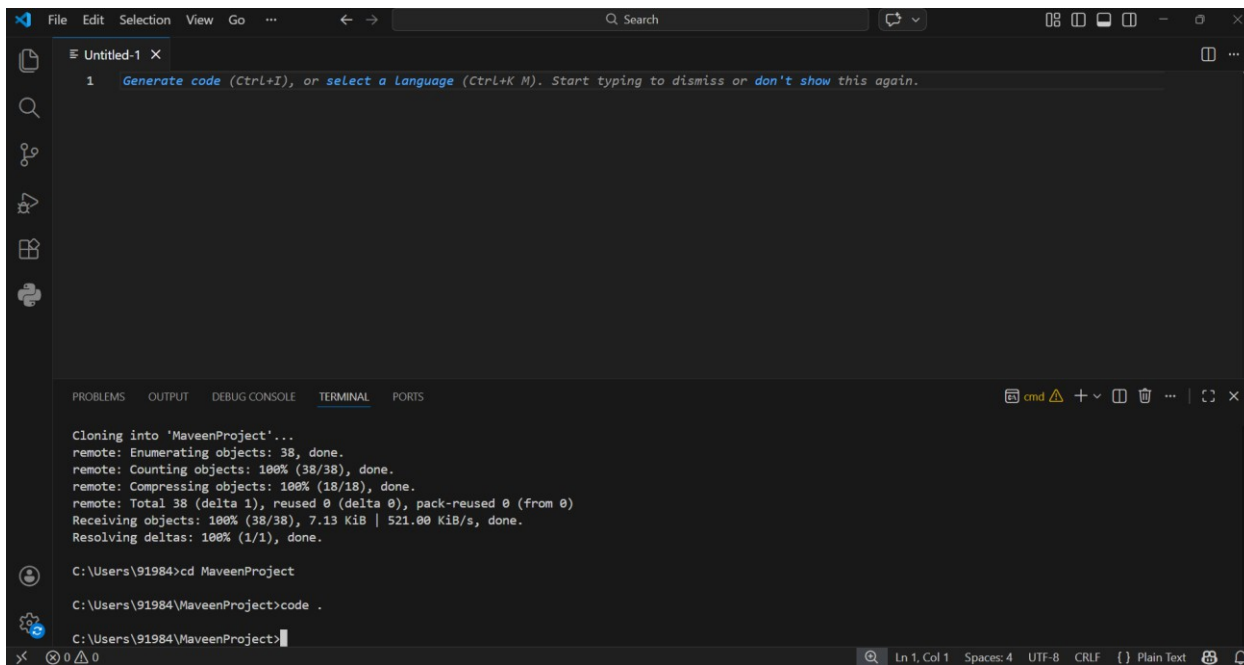
Write a corresponding unit test in AppTest.java.



```
1 package com.example;
2
3 import org.junit.jupiter.api.Test;
4 import static org.junit.jupiter.api.Assertions.*;
5
6 public class AppTest {
7
8     App calculator = new App();
9
10    @Test
11    void testAdd() {
12        assertEquals(15, calculator.add(10, 5));
13    }
14
15    @Test
16    void testSubtract() {
17        assertEquals(5, calculator.subtract(10, 5));
18    }
19
20    @Test
21    void testMultiply() {
22        assertEquals(50, calculator.multiply(10, 5));
23    }
24
25    @Test
26    void testDivide() {
```

3. Push and Clone the Repository

- Clone the repository using Visual Studio Code.

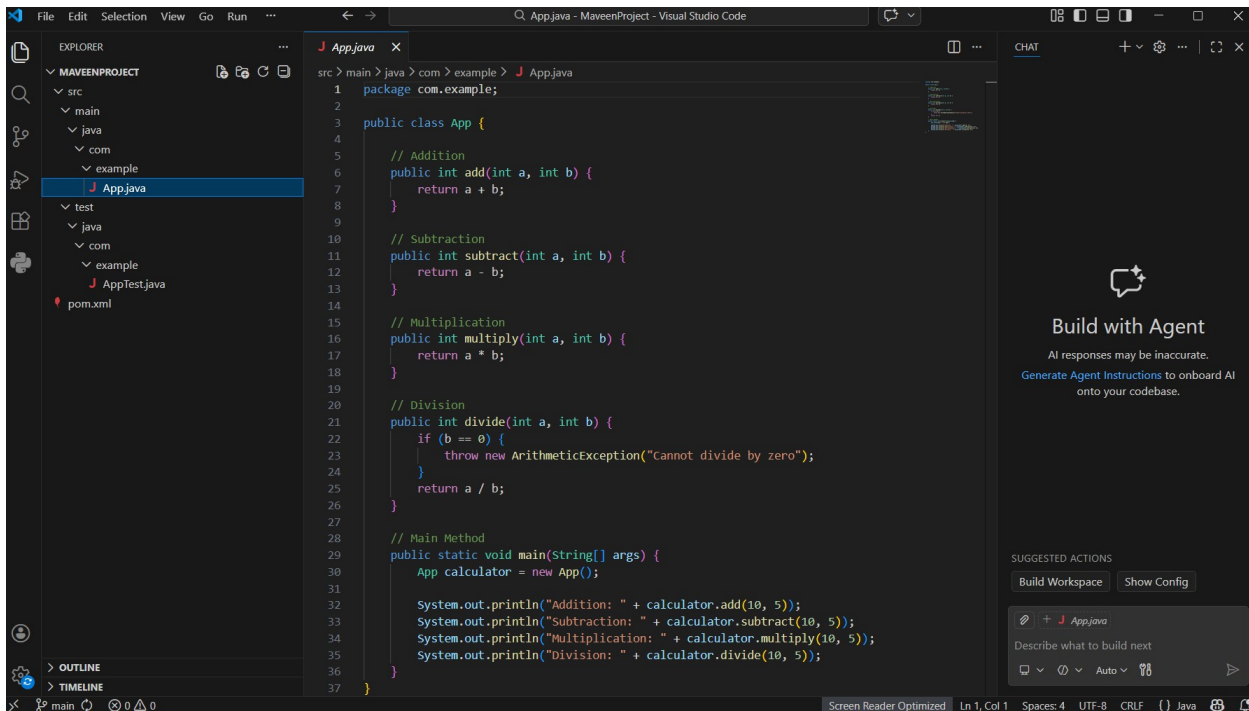


```
1 Generate code (Ctrl+I), or select a Language (Ctrl+K M). Start typing to dismiss or don't show this again.

Cloning into 'MaveenProject'...
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 38 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (38/38), 7.13 KiB | 521.00 KiB/s, done.
Resolving deltas: 100% (1/1), done.

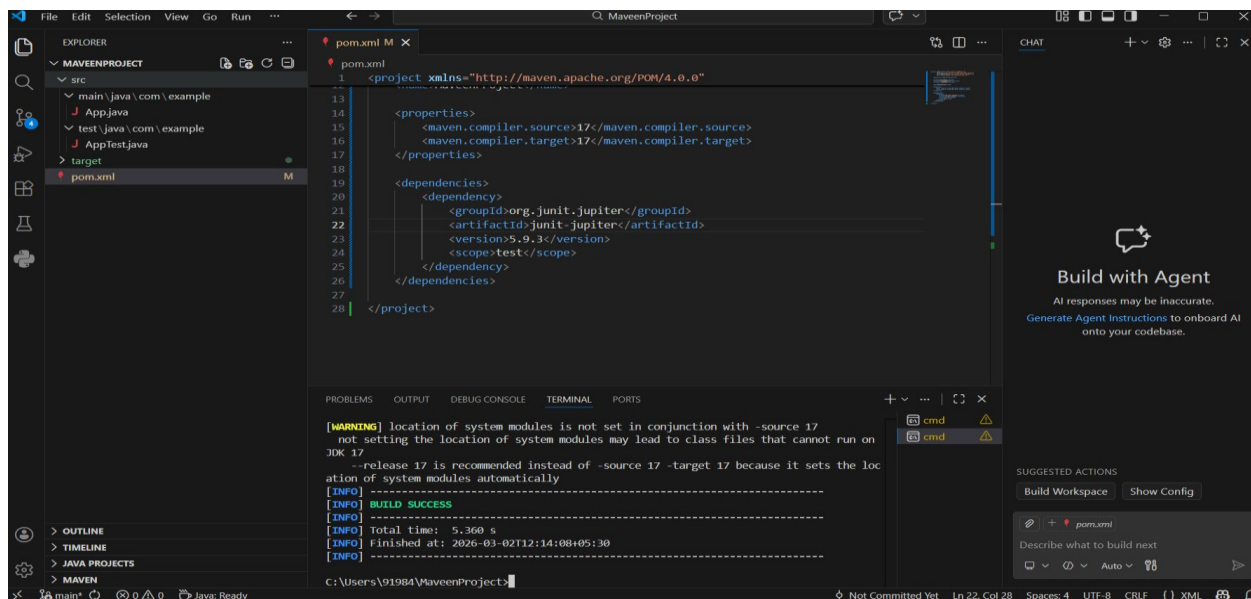
C:\Users\91984>cd MaveenProject
C:\Users\91984\MaveenProject>code .
C:\Users\91984\MaveenProject>
```

- Verify the project structure.

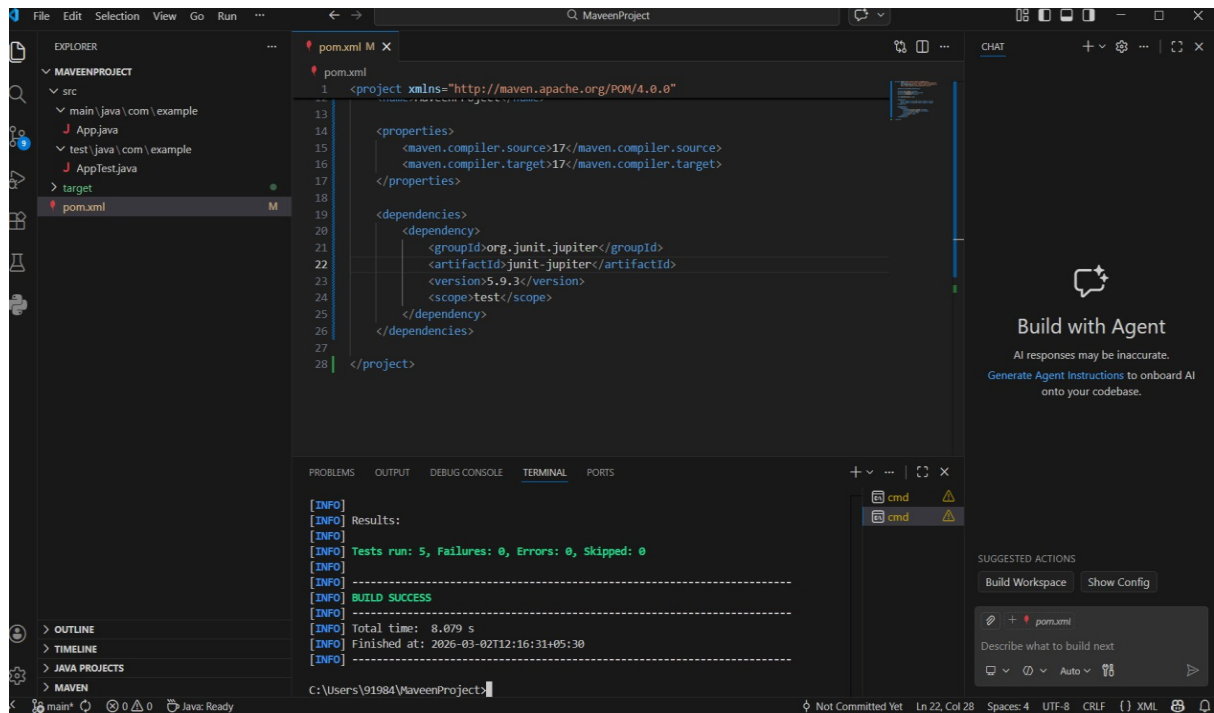


4. Build the Maven Project

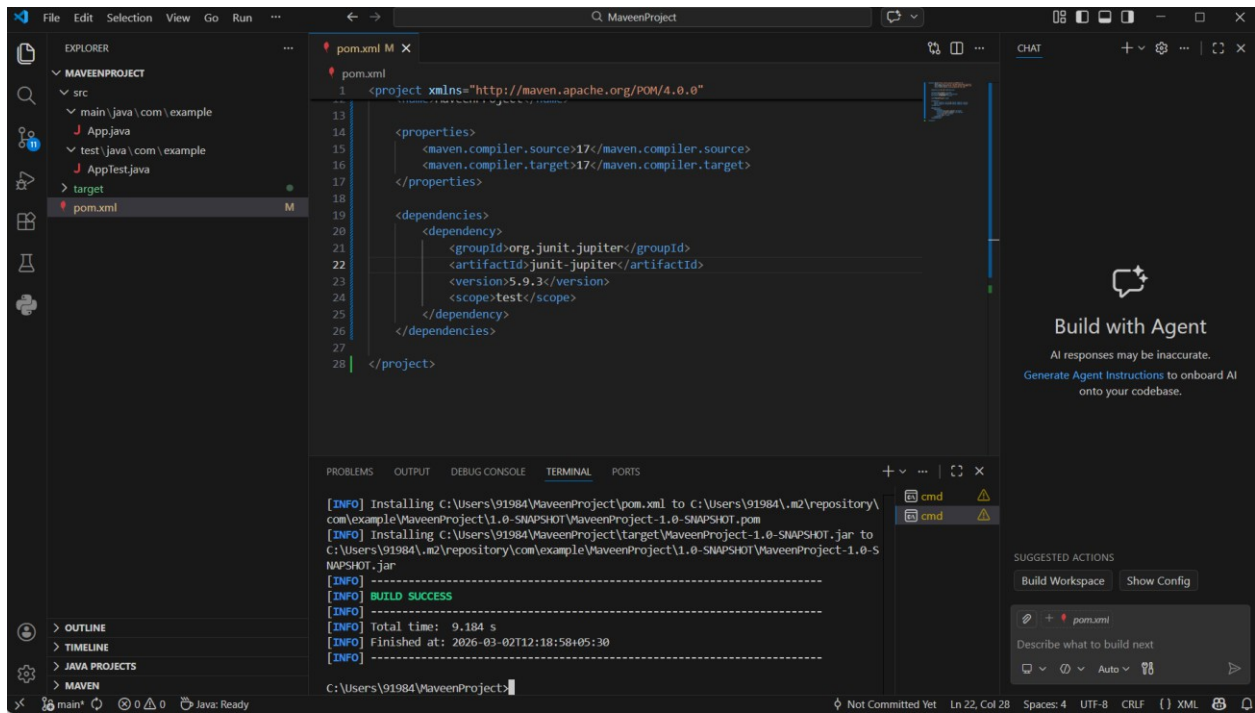
Use Maven commands in Visual Studio Code terminal to: Compile the project



Run tests



Build the project (mvn clean install)



5. Create and Run Jenkins Pipeline

Create a new Pipeline project in Jenkins.

 Jenkins

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

+

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	Git-Jenkins-Job2	21 days #35	21 days #33	1.6 sec
✓	☀	Job-A (Build)	21 days #1	N/A	0.16 sec
✓	☀	Job-B (Test)	21 days #1	N/A	0.14 sec
✓	☀	Job-C (Deploy)	21 days #1	N/A	0.13 sec
✓	☀	LK	21 days #1	N/A	0.29 sec
✓	☁	MavenProjectPipeline	2 min 20 sec #3	6 min 10 sec #2	1 min 16 sec
✓	☀	Pipeline-Demo	21 days #3	N/A	2.9 sec
✓	☀	Visual-Pipeline	21 days #3	N/A	1 sec

Create a pipeline script

 Jenkins

MavenProjectPipeline

Configure

General

Triggers

Pipeline

Advanced

Pipeline script

Script

```
1 pipeline {
2   agent any
3
4   tools {
5     maven 'Maven'
6     jdk 'JDK'
7   }
8
9   stages {
10
11     stage('Clone Repository') {
12       steps {
13         git branch: 'main',
14           url: 'https://github.com/laikishore2005-glitch/MaveenProject.git'
15       }
16     }
17   }
18 }
```

☒ Use Groovy Sandbox

Save

Apply

Execute the pipeline and verify successful build completion.

Jenkins / MavenProjectPipeline / #3

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

#3 (Mar 2, 2026, 12:37:05 PM)

Started by user Lalit Kishore

This run spent:

- 12 ms waiting;
- 1 min 16 sec build duration;
- 1 min 16 sec total from scheduled to completion.

git

Revision: 8a0d9ff57d1b4a5d0961ff33fa32fb98cc12f0bd
Repository: https://github.com/lalitkishore2005-github/MaveenProject.git

- refs/remotes/origin/main

Changes

1. Update pom.xml (details / githubweb)

REST API

Jenkins 2.541.1

Jenkins / MavenProjectPipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Builds >

Filter

Today

#3 12:37 PM

#2 12:33 PM

#1 12:28 PM

MavenProjectPipeline

Stage View

Average stage times:
(full run time: ~1min 16s)

	Declarative: Tool Install	Clone Repository	Compile	Test	Build
#3 Mar 02 12:37 1 commit	306ms	2s	2s	2s	26s
#2 Mar 02 12:33 No Changes	336ms	2s	Failed with the following error(s) See stage logs for more detail. Logs	157ms	145ms
#1 Mar 02 12:28 No Changes	5s	4s	311ms	187ms	149ms

Permalinks

- Last build (#2), 3 min 47 sec ago
- Last failed build (#2), 3 min 47 sec ago

THANK YOU